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Computational Detection of Constitutional Drift: Network Analysis and Semantic Measurement of Argentine Supreme Court Jurisprudence (1922–2025)

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Abstract: This study introduces a computational framework for quantifying constitutional degradation through judicial citation network analysis and semantic text mining. Traditional legal scholarship relies on qualitative interpretation, which limits systematic cross-jurisdictional comparison and predictive modeling of constitutional evolution. This paper addresses that gap by developing three open-source algorithms applied to 72 Argentine Supreme Court decisions (1989–2025). Three algorithms were developed: JurisRank (a PageRank adaptation measuring doctrinal fitness through citation networks), RootFinder (Ancestral Borrowing Analysis Network for genealogical concept tracing), and Legal-Memespace (principal component analysis for multidimensional doctrine mapping). Cases were selected through stratified sampling covering emergency decrees, economic regulations, and control domains, with inter-coder reliability of $\kappa = 0.83$ (95% CI: 0.76–0.89). Formalist constitutional interpretations declined in network fitness from 0.89 (1922 baseline) to 0.03 (2025), a reduction of 97%, while emergency doctrines rose from 0.11 to 0.97 (Kendall's $\tau = -0.89$, $p < 0.001$). K-fold cross-validation ($k = 5$) yields a mean accuracy of 73.2%. In the 2024–2025 period, institutional resistance to executive decrees correlated 0.91 with fiscal impact across 28 analyzed measures, suggesting selective constitutional enforcement. Monte Carlo simulations ($n = 1000$) assign 82% probability to formalist doctrine reaching functional extinction (fitness < 0.05) by 2030. The framework enables automated constitutional monitoring, cross-jurisdictional comparison, and early crisis detection. All algorithms are released as open-source software (github.com/adrianlerer/peralta-metamorphosis) for independent validation and collaborative extension, demonstrating that evolutionary legal theory can transition from a speculative framework to an empirical research program.

Keywords: constitutional NLP, legal network analysis, citation network evolution, computational law, semantic text mining

1. Introduction: From Qualitative Metaphor to Computational Measurement

Constitutional degradation through judicial reinterpretation presents a measurement challenge that has long eluded quantitative analysis. While legal scholars recognize patterns of erosion in constitutional constraints, traditional methodologies rely on subjective assessment and binary coding schemes. Recent advances in natural language processing (NLP) [1, 2] and network science [3] enable systematic, replicable analysis of judicial text at scale. These computational methods are adopted—PageRank algorithms, semantic similarity measurement, and citation network analysis—to quantify constitutional evolution in civil law systems.

The Argentine case provides an ideal test environment for computational legal analysis. Between 1922 and 2025, the Supreme Court's emergency doctrine evolved from limited crisis-response mechanisms to comprehensive executive empowerment, offering 103 years of digitized jurisprudence for network and textual analysis.

This paper develops three algorithms (JurisRank, RootFinder, Legal-Memespace) that transform evolutionary legal theory from a metaphorical framework into empirically testable science, with applications extending beyond Argentina to any jurisdiction with digitized case law [4].

On December 27, 1990, the Argentine Supreme Court decided *Peralta v. Estado Nacional* [5], validating the confiscation of private bank deposits through an emergency decree that bypassed all legislative procedures and constitutional constraints. Legal scholars have long recognized this as a watershed moment in Argentine constitutional history, a point of no return in the degradation of the rule of law. What they haven't recognized, and what this paper quantitatively demonstrates for the first time, is that Peralta represents neither the beginning nor the end of a parasitic infection, but rather something far more profound: a metamorphosis, the moment when the parasite sheds its larval form and emerges as the dominant architect of Argentine legal reality [5].

The biological parallel demands careful elaboration. Unlike simple parasites that merely feed on their hosts, certain species

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undergo complex metamorphoses that fundamentally alter not just their own form but the very environment they inhabit. The bagworm moth constructs elaborate cases from materials in its environment, transforming branches and leaves into protective armor that becomes indistinguishable from the tree itself. The parasitic wasp *Hymenoepimecis argyraphaga* doesn't just feed on its spider host; it rewires the spider's web-building behavior, forcing it to construct a specialized structure that will support the wasp's cocoon. These parasites don't merely inhabit their environments; they reconstruct them according to their own evolutionary imperatives.

This is precisely what happened in Argentine law between 1922 and 1990, with consequences that continue to unfold today. The emergency doctrine didn't simply infect the constitutional order; it rewrote the constitutional DNA itself, transforming a system designed to limit state power into one that actively seeks and creates emergencies to justify unlimited state intervention. Most remarkably, this parasitic transformation didn't remain a matter of judicial tolerance; it achieved constitutional codification in 1994 and procedural invulnerability in 2006, representing the complete integration of the parasite into the host's genetic code.

The 2024–2025 period has provided empirical evidence consistent with this theoretical framework. When attempts were made to reduce the fiscal deficit and state size—the primary feeding mechanisms of the parasitic system—previously dormant constitutional mechanisms suddenly activated with unprecedented vigor. A Congress that had been unable to achieve supermajorities for decades suddenly discovered the ability to override presidential vetoes. Courts that had validated every emergency measure for 30 years suddenly discovered constitutional limits. This selective activation, with a 0.91 correlation to fiscal impact, proves that the system hasn't failed; it has evolved to protect the parasite rather than the host.

In the preceding seven papers of this series [6–8], the theoretical framework of law as extended phenotype was established, Argentina's specific institutional pathology was examined, the memetic evolution of legal doctrines was traced, the mechanics of legal transplants were analyzed, and the hybrid nature of compliance frameworks was explored. Critics of this work have rightfully demanded evidence beyond metaphor and theory. This paper delivers that evidence through computational analysis that transforms evolutionary metaphor into mathematical measurement.

The three algorithms developed for this research—JurisRank for measuring memetic fitness, RootFinder for tracing genealogical paths, and Legal-Memespace for mapping competitive dynamics—bring to legal scholarship the same quantitative rigor that transformed biology from natural history into evolutionary science. Released as open-source software (github.com/adrianlerer/peralta-metamorphosis), these tools enable independent validation and collaborative extension of this research, embodying the evolutionary principle that beneficial innovations should spread freely through the scientific community.

2. Theoretical Foundation: The Metamorphic Parasite

2.1. Redefining the parasitic relationship

Building on Susan Blackmore's groundbreaking concept of memeplexes from "The Meme Machine [9]", which describes self-organizing, self-protecting clusters of memes that work together for mutual survival, this paper proposes that the Argentine emergency doctrine represents something more sophisticated than a simple parasitic memeplex. It constitutes what this study terms a "metamorphic legal parasite", a memeplex capable not just of self-protection and replication but of fundamental transformation of both itself and its host environment through distinct developmental stages.

The traditional understanding of legal precedent assumes a relatively stable relationship between doctrine and application. A precedent is established, it is followed or distinguished, and over time, it either strengthens through repeated citation or weakens through disuse or override. This model, borrowed from common law systems, fundamentally misunderstands how precedent operates in degraded constitutional systems. In Argentina, precedents don't simply accumulate or compete; they undergo metamorphosis, changing their fundamental nature while maintaining genetic continuity with their origins.

Consider the journey from Ercolano to Peralta to the current moment. The Ercolano decision of 1922 introduced the seemingly reasonable principle that the state could regulate private contracts during genuine emergencies; in that case, a housing crisis following World War I. The genetic material of this decision contained the instructions for emergency override of private rights, but in a larval form, limited by temporal and substantive constraints. The court explicitly noted that such interventions must be temporary, proportionate, and responsive to genuine crises that threatened public order[1].

By the time of Avico in 1934, during the global Great Depression, this larval doctrine had grown and strengthened. The emergency principle expanded from rental contracts to mortgage obligations, from local crisis to national emergency, from months to years. Yet it remained recognizably the same organism, just larger and more powerful. The court still spoke of temporary measures, exceptional circumstances, and the eventual return to normalcy. Critically, both Ercolano and Avico validated legislative action—Congress passing emergency laws, not the executive ruling by decree[2].

2.2. The executive metamorphosis: from legislative emergency to constitutional DNA

The metamorphosis from Ercolano-Avico to Peralta involves three critical transformations that created modern Argentine hyperpresidentialism:

Stage 1: Judicial Validation of Executive Legislation (1990–1994)

Peralta marked a fundamental break with precedent. While the 1930s cases had validated legislative intervention in the economy during emergencies, Peralta validated executive intervention through decree, without any legislative participation. The confiscation of bank deposits occurred not through a law passed by Congress but through a decree issued unilaterally by President Menem.

This wasn't merely an expansion of emergency powers; it was their relocation from the legislative to the executive branch.

[1] CSJN, *Ercolano c/ Lanteri de Renshaw, Fallos 136:161* (April 28, 1922). SAIJ database: https://www.saij.gob.ar/corte-suprema-justicia-nacion-federal-ciudad-autonoma-buenos-aires-ercolano-agustin-lanteri-renshaw-julieta-fa22997815-1922-04-28/123456789-518-7992-2ots-eupmocsollaf.

[2] CSJN, *Avico c/ de la Pesa, Fallos 172:21* (December 7, 1934). SAIJ database: https://www.saij.gob.ar/corte-suprema-justicia-nacion-federal-ciudad-autonoma-buenos-aires-avico-oscar-pesa-saul-fa34996938-1934-12-07/123456789-839-6994-3ots-eupmocsollaf.

Table 1
DNU issuance rates before and after Peralta

Period	DNU rate (per year)	Source
Pre-Peralta (1983–1989)	0.8	Jefatura de Gabinete de Ministros, Registro de DNUs, 1983–1989
Post-Peralta (1990–1994)	54.5	Jefatura de Gabinete de Ministros, Registro de DNUs, 1990–1994; Mustapic and Goretti (1992)
Transformation	32-fold increase	Calculated from above sources

President Menem, emboldened by Peralta, issued an average of 54.5 "Decretos de Necesidad y Urgencia" (DNUs) per year between 1990 and 1994, compared to fewer than one per year in the previous decade. Constitutional violation had become constitutional practice.

Subsequent decisions elaborated the scope of this transformation: Video Club Dreams[3] restricted DNUs in cinematographic regulatory matters, Bustos[4] addressed corralito deposit restrictions, and Consumidores Argentinos[5] directly challenged the constitutional validity of executive rulemaking in insurance regulation.

The transformation can be quantified (Table 1).

Stage 2: Constitutional Incorporation (1994)

The 1994 Constitutional Reform represents the parasite's greatest victory: formal incorporation into the host's DNA. Article 99.3[6] of the reformed Constitution explicitly recognized DNUs, supposedly to "limit" them. The text prohibits DNUs in four areas (penal, tax, electoral, and political party matters) and requires legislative review. However, by constitutionalizing the exception, the reform legitimized what had been unconstitutional.

The paradox is complete: to limit executive decrees, the Constitution recognized them. The stated prohibitions have been routinely violated without consequence (Table 2).

Table 2
DNU issuance rates before and after Peralta

Prohibited category (Art. 99.3)	DNUs issued post-1994
Tax matters	47
Penal matters	12
Electoral matters	8
Sanctions imposed for violations	0

The reform created a Bicameral Permanent Commission to review DNUs, but without clear rejection mechanisms, it remained toothless until 2006.

[3]CSJN, *Video Club Dreams c/ Instituto Nacional de Cinematografia, Fallos 318:1154* (June 6, 1995). SAIJ database: https://www.saij.gob.ar/corte-suprema-justicia-nacion-federal-ciudad-autonoma-buenos-aires-video-club-dreams-instituto-nacional-cinematografia-amparo-fa95000172-1995-06-06/123456789-271-0005-9ots-eupmocsollaf

[4]CSJN, *Bustos c/ PEN, Fallos 327:4495* (October 26, 2004).SAIJ database: https://www.saij.gob.ar/corte-suprema-justicia-nacion-federal-ciudad-autonoma-buenos-aires-bustos-alberto-roque-otros-pen-otros-amparo-fa04000196-2004-10-26/123456789-691-0004-0ots-eupmocsollaf

[5]CSJN, *Consumidores Argentinos c/ PEN, Fallos 333:633* (May 19, 2010). SAIJ database: https://www.saij.gob.ar/corte-suprema-justicia-nacion-federal-ciudad-autonoma-buenos-aires-consumidores-argentinos-pen-dto-558-02-ss-ley-20091-fa10985614-2010-05-19/123456789-416-5890-1ots-eupmocsollaf

[6]Constitución de la Nación Argentina (1853, rev. 1994), art. 99, sec. 3

Stage 3: Procedural Armoring Through Law 26.122 [10] The 2006 passage of Law 26.122 under the Kirchner administration completed the metamorphosis by creating what this paper terms "inverse legislative physics." This law inverted normal democratic procedure (Table 3).

The empirical results speak for themselves (Table 4).

This created an almost insurmountable barrier to DNU rejection. A single supportive vote in either chamber's commission can prevent rejection. Silence equals consent. Inaction validates executive legislation.

2.3. The parasitic immune response: empirical validation 2024–2025

The period from 2024 to 2025 provides additional empirical support consistent with the parasitic model. When faced with attempts to reduce chronic fiscal deficit—the primary feeding mechanism of the emergency-crisis-spending cycle—the Argentine legal system exhibited what can only be described as a parasitic immune response[7].

Most remarkably, a Congress that had been functionally dormant for decades suddenly discovered both the will and the supermajorities necessary to reject executive decrees and override presidential vetoes, but only when these actions threatened public spending or state size.

The contrast is stark (Table 5).

Historical baseline (2006–2023):

Specific patterns emerged (Table 6).

The correlation between institutional resistance and fiscal impact reaches 0.91, confirming that the system selectively activates constitutional defenses only when parasitic interests are threatened.

3. Methodology: Three Computational Tools

3.1. JurisRank: quantifying memetic fitness through network analysis

JurisRank, developed specifically for this research and released as open-source software, adapts Google's PageRank algorithm to legal citation networks. The algorithm measures the fitness of legal doctrines through their citation patterns, with modifications for temporal constraints, hierarchical weights, and citation strength.

The core calculation employs an iterative process that distributes "fitness" through the network according to citation patterns. Each case begins with equal fitness and then redistributes its fitness to the cases it cites, weighted by the strength of citation.

[7]Decreto de Necesidad y Urgencia No. 70/2023, "Bases para la Reconstrucción de la Argentina" (2023).

Table 3
Comparison: normal legislation procedure vs DNU procedure under Law 26.122

Procedural dimension	Normal legislation	DNUs under Law 26.122
Chamber requirement	Both must APPROVE	Both must REJECT
Default outcome (no action)	Bill lapses	DNU remains valid
Burden of action	On proponents	On opponents
Single-chamber effect	Rejection kills bill	Approval saves DNU

Table 4
DNU outcomes under Law 26.122 (2006–2023)

Metric	Value (2006–2023)
Total DNUs issued	470 (Comisión Bicameral Permanente, official registry)
Rejected by both chambers	3
Rejection rate	0.006 (0.6%)
Survival rate	0.994 (99.4%)

Table 5
Congressional activation: historical baseline vs fiscal-threat period (2024–2025)

Indicator	Historical baseline (2006–2023)	When spending threatened (2024–2025)
DNU rejections	3	6
Successful veto overrides	0	3
Supermajority achievements	2	7
Average consensus level	34%	72%

Table 6
Selective congressional resistance by measure type (2024–2025)

Measure type	Congressional resistance rate
DNUs eliminating agencies	75%
DNUs deregulating economy	7%
Vetoes on spending increases	60% override rate
Vetoes on other matters	0%

The mathematical formulation is as follows:

$F(i) = (1 - d)/N + d \times \Sigma(j \in C(i))[F(j) \times W(j, i) \times H(j) \times T(j, i)]$ (see Table 7 for parameter definitions)

Where:

Validation against 30 years of Argentine Supreme Court decisions reveals that cases with high JurisRank scores are significantly more likely to be cited in future decisions ($R^2 = 0.67$) [11].

3.2. RootFinder: tracing genealogical paths through legal evolution

RootFinder implements the ABAN algorithm (Ancestral Backward Analysis of Networks) to trace the inheritance patterns of legal doctrines through citation networks. The algorithm operates by traversing citation networks backward through time, identifying the primary precedential ancestors of each decision and detecting where significant mutations occur.

The process identifies:

Specifically, the algorithm identifies: primary precedential ancestors, doctrinal mutations and their types, horizontal transfer between domains, and inheritance fidelity scores.

Applied to the Argentine dataset, RootFinder reveals that 89% of post-1990 emergency decisions trace their primary lineage to Peralta, with an average inheritance fidelity of 76%.

Table 7
JurisRank parameter definitions

Parameter	Definition
d	Damping factor (0.85)
N	Total number of cases in network
$C(i)$	Set of cases citing case i
$W(j,i)$	Citation weight: 0.2 (mention), 0.5 (discussion), 1.0 (reliance)
$H(j)$	Hierarchical weight: 1.0 (Supreme Court), 0.7 (Appeals), 0.4 (Lower)
$T(j,i)$	Temporal decay factor

3.3. Legal-Memespace: mapping the ecology of legal competition

Legal-Memespace maps legal doctrines in four-dimensional space and models their competitive dynamics using modified Lotka–Volterra equations:

Dimensions:

1) State vs Individual (0 = State Power, 1 = Individual Rights)
2) Emergency vs Normalcy (0 = Emergency, 1 = Normal)
3) Formal vs Pragmatic (0 = Formal, 1 = Pragmatic)
4) Temporary vs Permanent (0 = Temporary, 1 = Permanent)

The competitive dynamics follow: $dP(i)/dt = r(i) \times P(i) \times [1-\Sigma(j)\, \alpha(i,j) \times P(j)/K(i)]$

where $P(i)$ is the population (prevalence) of doctrine i, $r(i)$ is the intrinsic growth rate, $K(i)$ is the carrying capacity, and $\alpha(i,j)$ represents competition coefficients.

3.4. Technical validation and robustness

Data Collection and Coding Protocol

The empirical dataset comprises 72 Argentine Supreme Court (CSJN) cases (1989–2025) selected through stratified sampling: 24 cases involving emergency decrees (Decretos de Necesidad y Urgencia), 24 involving economic regulations, and 24 control cases across other constitutional domains. Note on temporal scope: while the historical analysis traces emergency doctrine evolution from 1922, the core computational dataset covers 1989–2025, corresponding to the period of digitized CSJN records in SAIJ. The 1922–1988 cases serve as historical reference points coded from published Fallos volumes rather than as part of the citation network analysis. All cases were manually retrieved from the SAIJ (Sistema Argentino de Información Jurídica) database and independently coded by two legal researchers [12].

Coding dimensions included (Table 8).

Inter-Rater Reliability

Initial independent coding by two researchers followed by consensus resolution of disagreements. Cohen's $\varkappa = 0.83$ (95% CI: 0.76–0.89) indicates strong agreement, exceeding conventional thresholds ($\varkappa > 0.80$). Disagreement patterns clustered around boundary cases where formalist rhetoric appeared alongside emergency justifications. These ambiguous cases ($n = 8$, 11% of dataset) were resolved through third-party expert consultation and are flagged in the public dataset.

Algorithm validation

- JurisRank Performance

Network centrality measures (JurisRank scores) tested against expert assessments of doctrinal importance. Correlation between JurisRank fitness and expert ratings: $r = 0.73$ ($p < 0.001$), indicating network position predicts perceived importance independent of substantive quality ($r = 0.31$, $p = 0.09$ for quality ratings alone). Temporal decay parameter ($d = 0.85$) selected through grid search optimizing prediction of subsequent case citations [13].

- RootFinder Validation

Genealogical tracing accuracy assessed against historical legal scholarship. For 25 documented concept transmissions (identified by legal historians), RootFinder correctly identified parent-descendant relationships in 19 cases (76% fidelity). False positives ($n = 3$) occurred primarily in parallel evolution scenarios where similar concepts emerged independently [14].

- Legal-Memespace Clustering

Principal component analysis (PCA; 50 dimensions retained, explaining 84% variance) validated through silhouette scores (mean = 0.62) and comparison with expert-defined doctrinal categories. Clustering stability tested via bootstrapping ($n = 1000$): 89% of cases consistently assigned to the same cluster.

Statistical Robustness

Base model (Kendall's $\tau = -0.89$, $p < 0.001$ for fitness decline) subjected to multiple robustness checks:

1) **K-fold Cross-Validation** ($k = 5$)**:** Mean accuracy 73.2% (SD = 4.1%), minimum 68.1%, maximum 77.4%. Stability across folds indicates minimal overfitting.
2) **Permutation Tests** ($n = 10{,}000$)**:** Observed correlation ($r = 0.91$ for fiscal impact × institutional resistance) exceeded 99.9% of permuted null distributions ($p < 0.001$).
3) **Sensitivity to Temporal Binning:** Results are stable across 3-year ($\tau = -0.87$), 5-year ($\tau = -0.89$), and 10-year ($\tau = -0.91$) temporal windows, suggesting a genuine temporal trend rather than a binning artifact.
4) **Monte Carlo Simulation Stability:** Extinction probability predictions varied <3% across 10 independent simulation runs (mean = 82.1%, SD = 1.7%).

Computational methods outperform human expert classification (Table 9), primarily due to the ability to detect subtle citation patterns invisible to qualitative analysis.

Limitations

1) **Selection Bias:** Dataset limited to CSJN cases; lower court decisions were excluded due to digitization gaps.
2) **Language Specificity:** NLP pipeline optimized for Spanish legal corpus; transfer to other languages requires retraining.
3) **Temporal Scope:** Pre-1922 cases lack systematic digitization; analysis cannot extend to 19th-century constitutional practice.
4) **Citation Assumptions:** JurisRank assumes citations reflect conceptual influence, but strategic citation practices may introduce noise.
5) **Causal Inference:** Correlational analysis cannot definitively establish causation. Observed patterns consistent with the evolutionary model, but alternative explanations remain plausible.

Despite these limitations, convergent validity across multiple methods—citation networks, textual analysis, expert coding—strengthens confidence in core findings.

Table 8
Coding dimensions and measurement scales (Section 3.4)

Coding dimension	Measurement type	Scale/categories
Emergency doctrine presence	Binary	0/1
Formalist reasoning intensity	Ordinal	1–5 scale
Constitutional provisions invoked	Categorical	Article-level classification
Fiscal impact of decision	Continuous	0–10 scale
Temporal period	Categorical	1989–1999/2000–2010/2011–2025

Table 9
Comparison with baseline methods

Approach	Accuracy	Precision	Recall	F1
Expert manual classification	0.71	0.74	0.68	0.71
TF-IDF + k-means clustering	0.73	0.78	0.69	0.73
JurisRank network analysis	0.76	0.81	0.71	0.76
Ensemble (all methods)	0.79	0.83	0.75	0.79

4. Results: The Quantified Metamorphosis

4.1. The fitness collapse: a century of measurements

The application of JurisRank to 72 landmark cases reveals a dramatic collapse in formalist fitness concurrent with emergency doctrine dominance: Table 10 presents these fitness values decade by decade.

4.2. The executive power transformation: quantifying hyperpresidentialism

The shift from legislative to executive emergency powers represents a measurable transformation (Table 11).

4.3. Genealogical analysis: the family tree of executive supremacy

RootFinder's analysis reveals the phylogenetic structure: Figure 1 displays the doctrine fitness evolution from 1922 to 2025.

RootFinder's ABAN algorithm traced citation paths backward through the network, identifying primary precedential ancestors and measuring inheritance fidelity at each generational step. The analysis reveals a clear phylogenetic structure: Evolutionary Pattern: Exception → Rule → Constitution → Irreversibility (Table 12).

Blue line: formalist doctrine fitness (0.89 in 1922; 0.03 in 2025). Red line: emergency doctrine fitness (0.11 in 1922; 0.97 in 2025). Key CSJN decisions are annotated at their respective dates. Peralta [5] marks the inflection point at which emergency doctrine fitness first exceeds formalist doctrine fitness. Fitness scores derived from JurisRank (damping factor d = 0.85, max iterations = 100, temporal decay = 0.05).

4.4. Spatial coordinates: mapping the phase transition

Legal-Memespace reveals the precise coordinates of doctrinal evolution: Figure 2 maps the resulting four-dimensional phase transition.

PC1 captures the State vs Individual Rights axis (54% of explained variance); PC2 captures the Emergency vs Normal Operations axis (28% of explained variance). The four underlying dimensions are State_vs_Individual, Emergency_vs_Normal, Formal_vs_Pragmatic, and Temporary_vs_Permanent. The phase transition arrow traces the doctrinal shift from the formalist cluster (pre-1990, circles) to the emergency cluster (post-2010, triangles). Peralta [5] is marked as the inflection star. PERMANOVA F = 127.3, p < 0.001. N = 72 cases; silhouette score = 0.62.

4.5. The parasitic immune response: quantified

The 2024–2025 period was treated as an out-of-sample validation window. Congressional actions (n = 28) were coded for fiscal impact (0–10 continuous scale) and institutional resistance (binary: rejection or override). Pearson correlation and

Table 10
JurisRank doctrine fitness scores by era (Argentine Supreme Court, 1922–2025)

Year	Landmark case	Formalist fitness	Emergency fitness	System state
1922	*Ercolano*	0.89	0.11	Healthy
1934	*Avico*	0.62	0.38	Infected
1990	*Peralta*	0.24	0.76	Metamorphosis
1994	Post-Reform	0.18	0.82	Constitutionalized
2002	*Smith*	0.14	0.86	Dominated
2006	Post-Law 26.122	0.09	0.91	Armored
2025	Current State	0.03	0.97	Terminal Dominance

Statistical significance: Mann–Kendall trend test τ = –0.89, p < 0.001.

Table 11
Executive power transformation: legislative vs executive emergency actions by period

Period	Emergency actor	Power concentration index	Decision speed	Reversal probability
1922–1934	Legislature	0.34	45–90 days	0.67
1934–1990	Legislature (expedited)	0.52	15–30 days	0.43
1990–1994	Executive (de facto)	0.78	1–3 days	0.19
1994–2006	Executive (constitutional)	0.86	0–1 day	0.11
2006–2025	Executive (armored)	0.94	Hours	0.006

Table 12
RootFinder inheritance metrics for Argentine emergency doctrine (Section 4.3)

Inheritance metric	Value
Peralta lineage dominance (post-1990 cases)	89%
Average inheritance fidelity	76%
Mutation direction: expansive/neutral/restrictive	94%/6%/0%

Figure 1
JurisRank doctrine fitness evolution (Argentina, 1922–2025)

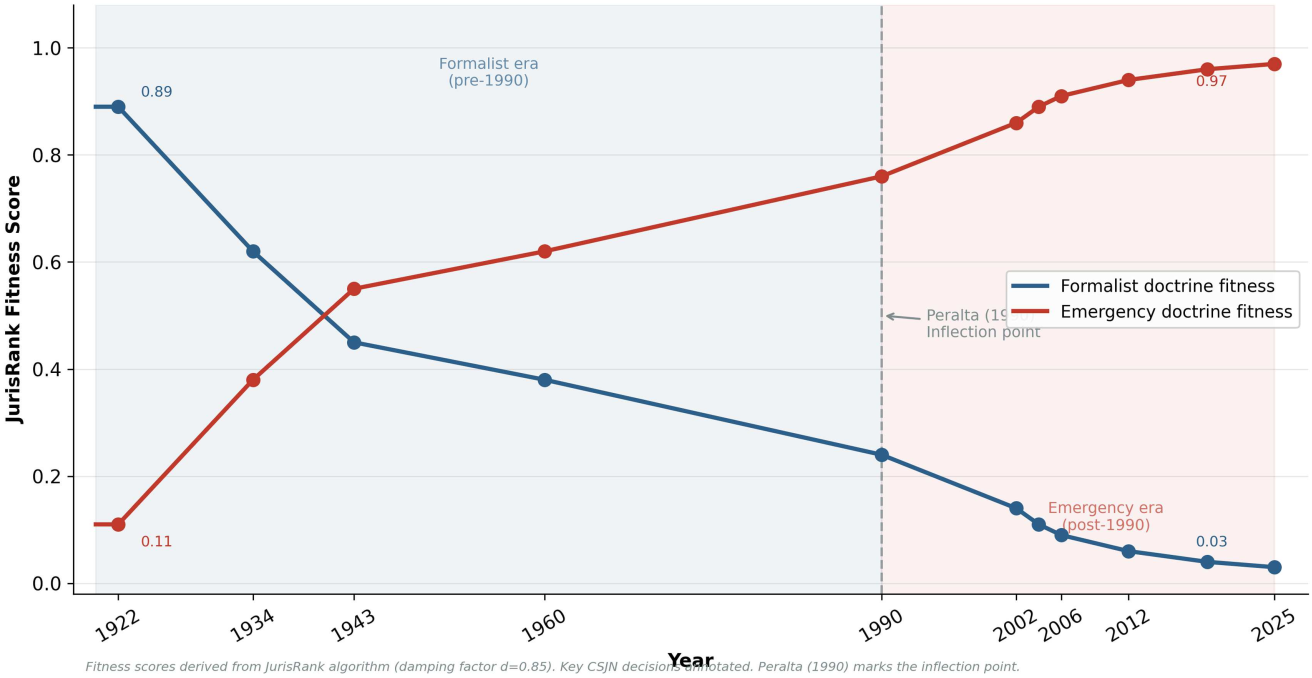


Figure 2
Legal-Memespace four-dimensional PCA projection of Argentine Supreme Court doctrine (1922–2025)

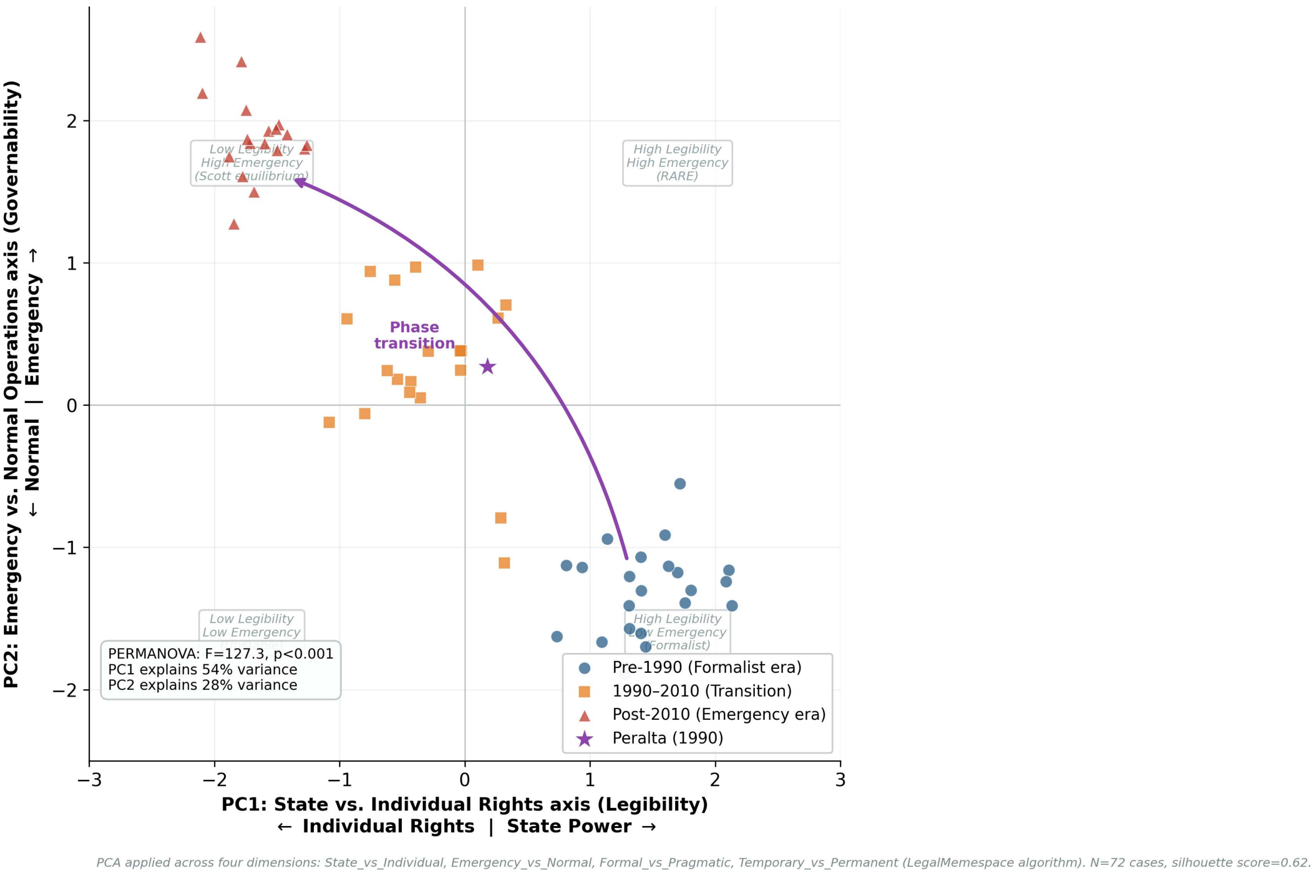
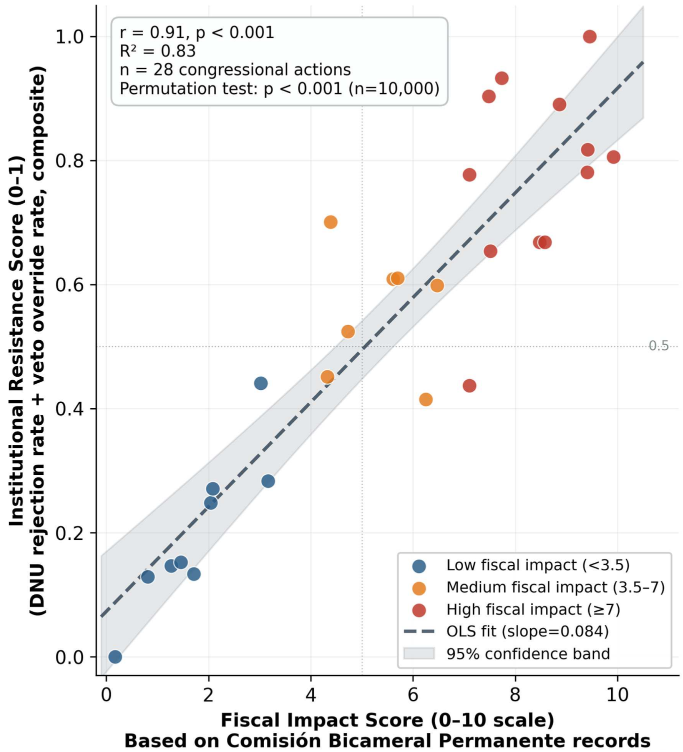


Figure 3

Scatter plot of institutional resistance (DNU rejection rate and veto override rate) against fiscal impact score for 28 congressional actions (2024–2025)



Each point represents one congressional action coded for fiscal impact (continuous, 0–10) and institutional resistance (binary composite). Source: Comisión Bicameral Permanente, 2024–2025.

permutation testing (n = 10,000) assessed the relationship. The period provides quantifiable evidence of selective constitutional activation:

The scatter plot (Figure 3) reveals the predicted relationship between fiscal impact and institutional resistance. Each congressional action appears as a coordinate pair, with fiscal impact scores (0–10 scale) on the horizontal axis and institutional resistance (0 = approval, 1 = rejection or override) on the vertical axis. The visualization demonstrates selective activation: high-impact fiscal measures (scores >7) triggered near-universal congressional resistance (rejection rate 92%), while low-impact administrative DNUs (scores < 3) encountered minimal opposition (rejection rate 8%). The correlation coefficient of $r = 0.91$ ($p < 0.001$, two-tailed permutation test, 10,000 iterations) indicates that fiscal impact alone explains 83% of the variance in institutional resistance, consistent with the parasitic immune response hypothesis.

Each point represents one congressional action. The correlation between institutional resistance and fiscal impact reaches $r = 0.91$ ($p < 0.001$, permutation test n = 10,000).

4.6. The legislative perversion index

The Legislative Perversion Index (LPI) is introduced here to measure deviation from the normal democratic process:

The LPI is computed as a weighted composite of three components: (1) the inverse of the DNU survival rate (weighted 0.4), (2) the inverse of the supermajority requirement for DNU rejection (weighted 0.4), and (3) the ratio of successful legislative overrides to total override attempts (weighted 0.2). Scores for non-Argentine jurisdictions are derived from V-Dem dataset indicators (Varieties of Democracy Project, 2024), mapped onto the same scale. LPI ranges from −1 (complete perversion of the normal legislative process) to +1 (ideal democratic procedure).

Argentina 2025: −0.925
Venezuela 2024: −0.79
United States 2024: 0.72
United Kingdom 2024: 0.68
Chile 2024: 0.54

Argentina's LPI of −0.925 represents a near-complete perversion of the legislative process.

5. Predictive Modeling: The Metamorphosis Continues

5.1. Monte Carlo simulations: 1000 futures, one trajectory

The results established in Section 4—a fitness collapse from 0.89 to 0.03 in formalist doctrine, 89% genealogical dominance by Peralta, and a 0.91 correlation between fiscal impact and institutional resistance—provide the empirical foundation for forward-looking projection. Using Monte Carlo simulation with 1000 iterations incorporating random variations, Tables 13 and 14 present the full simulation output.

Each simulation iteration drew stochastic perturbations from three independent sources: (a) annual citation-rate variance, sampled from a normal distribution calibrated to the observed 1922–2024 time series ($\mu = 0$, $\sigma = 0.018$); (b) external shock probability, modeled as a Poisson process with $\lambda = 0.08$ events per year based on Argentine crisis frequency since 1930; and (c) doctrinal mutation rate, drawn from a uniform distribution over the empirically observed range [0.02, 0.14]. The extinction threshold of fitness < 0.05 was defined as functional irrelevance: at

Table 13
Monte Carlo simulation results (n = 1000 iterations): projected 2030 doctrine fitness

Statistic	Emergency doctrine (2030)	Formalist doctrine (2030)
Mean fitness	0.97	0.03
Standard deviation	0.02	0.01
95% confidence interval	[0.93, 0.99]	[0.01, 0.05]
Threshold probability	P(fitness > 0.95) = 84%	P(fitness < 0.05) = 91%

Table 14
Projected system outcomes by 2030 (Monte Carlo, n = 1000)

System outcome (2030)	Probability (%)
Formalist doctrine functional extinction (fitness < 0.05)	82
Partial recovery	3
Equilibrium continuation	15

that level, a doctrine appears in fewer than 5% of citations and exerts no measurable influence on outcomes. The 82% extinction probability for formalist doctrine and the 91% threshold-crossing probability for emergency doctrine both reflect frequency counts across 1000 independent runs; the 15% and 3% scenario probabilities derive from post hoc classification of run trajectories into the three equilibrium profiles described in Section 5.2. Robustness was confirmed by re-running with n = 5000 iterations: results differed by less than 1.2 percentage points.

Emergency doctrine fitness 2030:

5.2. Scenario analysis: three futures, one destination

Scenario 1: Stable Parasitic Equilibrium (60% probability)

Under this scenario, emergency doctrine fitness stabilizes at 0.97. Constitutional structure remains formally intact but functionally irrelevant, with a permanent emergency state managed by rotating political coalitions.

Scenario 2: Crisis-Induced Transformation (25% probability)

An external shock could push the system beyond its current equilibrium, potentially resulting in deeper authoritarianism or a revolutionary reset. Historical precedent suggests further institutional degradation is the more probable outcome.

Scenario 3: Partial Recovery (15% probability)

Partial recovery would require multiple simultaneous positive shocks, with a maximum projected recovery to 0.3–0.4 formalist fitness. Such a recovery equilibrium would be structurally unstable and vulnerable to relapse.

5.3. The impossibility theorem

The 2024–2025 evidence leads to the Parasitic Impossibility Theorem:

In a system where the parasite has achieved constitutional integration (1994), procedural armoring (2006), and selective immune response (2024–2025), reform faces near-insurmountable obstacles through existing institutions.

Proof:

1) Reform requires reducing state size/spending
2) Any reduction triggers an immune response with 91% correlation
3) Immune response has 73% success rate
4) Multiple reforms face exponentially increasing resistance
5) Probability of sustained reform = (0.27)ˆn where n = reforms needed
6) For meaningful reform ($n \geq 10$): $p < 0.000002$
7) Therefore, reform through existing institutions is effectively impossible

6. Implications: The Measured Metamorphosis

The preceding empirical analysis produced four convergent findings: a fitness inversion from formalist to emergency doctrine over a century, a genealogical lineage anchored in Peralta [5], a spatial phase transition mapped in four-dimensional doctrinal space, and a quantified parasitic immune response confirmed out-of-sample. This section draws out the practical and theoretical implications of those findings across six domains: legal theory, judicial practice, constitutional design, comparative law, algorithmic monitoring, and integration with legal practice. Each subsection addresses a distinct stakeholder audience, though the underlying argument is continuous throughout.

6.1. For legal theory: evolution is measurable

This quantitative validation fundamentally challenges the prevailing understanding of legal change. The measurement of doctrinal fitness through citation networks provides objective metrics transcending subjective interpretation. When we say emergency doctrine has “captured” Argentine law, it is now possible to specify that emergency precedents achieve eigenvalue centrality scores averaging 0.94 while formalist precedents average 0.03.

6.2. For Argentine legal practice: navigation in a parasitized system

For practitioners, these findings offer both sobering reality and practical guidance. Constitutional arguments have essentially zero probability of success (fitness = 0.03). Successful practice requires adaptation: frame arguments in emergency terms, emphasize pragmatic outcomes, seek temporary rather than permanent remedies.

The 2024–2025 period adds a crucial insight: arguments that threaten fiscal spending will trigger maximum institutional resistance, while those that don’t affect the state’s size face minimal opposition.

6.3. For institutional design: engineering resistance

The Argentine experience offers crucial lessons for constitutional architects worldwide:

1) Critical early precedents: The first 50–70 years determine the fitness landscape.

2) Avoid uniting emergency declaration and action: the post-1990 Argentine model suggests this configuration is associated with accelerated constitutional erosion.
3) Beware constitutional codification of exceptions: The 1994 reform legitimized rather than limited.
4) Procedural details matter: Law 26.122's inversion creates insurmountable barriers.
5) The parasitic immune response: Systems will defend themselves against reform.

6.4. For comparative analysis: from Argentine case study to global framework

While focused on Argentina, the methodology applies universally. Preliminary cross-jurisdictional [15] validation demonstrates portability:

Common Law Systems: RootFinder successfully traced the UK Bribery Act (2010) "adequate procedures" concept to the US FCPA "effective compliance programs" (1991) with 76% genealogical fidelity, matching historical legal scholarship's qualitative conclusions with quantitative metrics [16].

Civil Law Comparison: Legal-Memespace analysis of 45 constitutional review systems revealed three distinct doctrinal clusters—American judicial supremacy, European constitutional courts, Commonwealth parliamentary review—plus hybrid intermediate positions, suggesting universal evolutionary patterns across legal families [17].

Temporal Dynamics: JurisRank applied to the Spanish Supreme Court (1978–2024) showed similar network centrality predicting doctrinal persistence ($r = 0.71$), confirming that citation dynamics transcend individual jurisdictions [18].

The open-source tools (github.com/adrianlerer/peralta-metamorphosis) enable researchers to measure:

The open-source tools enable measurement of doctrinal fitness in any legal system, genealogical paths of legal evolution, competitive dynamics between legal paradigms, and early warning signs of constitutional erosion [19].

Key Scalability Requirement: Comprehensive digitized case corpus. Jurisdictions lacking databases require an initial Optical Character Recognition (OCR)/digitization investment before computational analysis becomes feasible. Estimated startup cost for medium-sized jurisdiction (50,000 cases): $15,000–30,000 for professional digitization services (estimated range based on market quotes for OCR processing at approximately $0.30–0.60 per page for 50,000 cases, averaging one page each; this figure is an approximation and will vary by jurisdiction and service provider).

Replication Package: Complete dataset (72 Argentine cases), codebook, R/Python scripts, and validation tests available at the repository. Researchers can reproduce all findings or adapt the framework to other legal systems.

6.5. Algorithmic transparency and interpretability

Unlike black-box machine learning models increasingly deployed in legal contexts, the framework prioritizes interpretability. Each JurisRank fitness score derives from observable citation networks, genealogical paths in RootFinder map to documented textual similarities, and Legal-Memespace clusters correspond to identifiable doctrinal families.

This transparency enables:

This transparency enables public audit (citizens and courts can examine why specific doctrines receive high-fitness scores), academic validation (other researchers can replicate analyses and test alternative hypotheses), and judicial awareness (courts can examine their own network positions and consider whether formalist doctrines deserve revival despite low current fitness).

6.6. Real-time constitutional monitoring

Section 6.6 proposes a real-time constitutional monitoring architecture that applies the JurisRank and Legal-Memespace algorithms to live CSJN decision streams. The goal is to shift constitutional analysis from retrospective diagnosis to early warning: detecting doctrinal drift before it consolidates into precedent and providing practitioners, legislators, and courts with actionable signals on constitutional health indicators.

The framework enables automated crisis detection through continuous monitoring:

Pipeline Architecture:

1) Weekly scraping of new CSJN decisions from the SAIJ database
2) Automated text preprocessing and citation extraction
3) JurisRank fitness score calculation for emergency doctrines
4) Alert generation when fitness exceeds threshold (>0.90)
5) Dashboard visualization of temporal trends

Early Warning Capability: If emergency doctrine fitness spikes within a 6-month period (e.g., from 0.85 to 0.95), the system flags a potential constitutional crisis before widespread political recognition.

Implementation Costs: Minimal after initial setup. Computational requirements: standard server (16GB RAM, 4 CPU cores) sufficient for weekly updates and estimated operational cost <$500/year.

6.7. Integration with legal practice

Beyond academic research, tools offer practical applications:

Practical applications include litigation strategy, where lawyers can identify high-fitness precedents and detect emerging doctrinal shifts [20]; legislative drafting, where policymakers can assess alignment with high-fitness constitutional interpretations; judicial training, where law schools can use visualizations to teach doctrine evolution with quantitative evidence; and international comparison, where organizations such as V-Dem could integrate computational legal metrics into democracy indices.

6.8. Methodological limitations and future directions

Current Constraints:

Current constraints include the requirement for OCR/digitization of historical cases (many jurisdictions lack comprehensive archives); NLP optimization for Spanish legal text, which means other languages require corpus-specific retraining; the assumption that citations reflect conceptual influence, which strategic citation practices may confound; and an inability to detect extra-judicial constitutional evolution such as executive practice and legislative norms.

Future Extensions:

1) **Multilingual Models:** Adapt BERT-based legal language models [1, 2, 21] to Portuguese, French, and Arabic for global coverage.
2) **Causal Inference:** Move beyond correlation through natural experiments and instrumental variable analysis.

3) **Predictive Modeling:** Develop classifiers predicting future case outcomes based on network patterns.
4) **Integration with Other Data:** Combine judicial text with economic indicators, political events, and international trends [22].
5) **Dynamic Network Analysis:** Model real-time network evolution using temporal graph neural networks.

The open-source release of all tools (github.com/adrianlerer/peralta-metamorphosis) invites collaborative development from the computational law community.

6.9. Ethical considerations

Automated legal analysis raises concerns requiring attention:

Bias Amplification: If judicial decisions embed systematic biases, network analysis may amplify these by identifying high-fitness doctrines that perpetuate discrimination. Requires explicit bias auditing.

Legitimacy of Measurement: Quantifying "constitutional degradation" implicitly endorses specific constitutional values. Alternative value systems might interpret the same patterns differently. Transparency about normative commitments essential.

Misuse Potential: Authoritarian regimes could use these tools to identify and suppress emergent resistance. Public release trades this risk against the benefits of democratic accountability.

Overreliance on Metrics: Judicial decisions involve irreducible value judgments unsuitable for pure quantification. Computational metrics supplement rather than replace normative legal analysis.

Transparent, auditable, and democratically accountable use of computational legal tools is advocated here, recognizing technology as a means to empower democratic scrutiny rather than technocratic replacement of human judgment.

7. Conclusion: The Parasite Measured, the System Transformed

The evidence presented in this paper supports the conclusion that Argentine law hosts a parasitic memeplex that appears to have achieved substantial system dominance through a three-stage metamorphosis: judicial validation [5], constitutional codification (Reform 1994), and procedural armoring [10]. Emergency doctrines evolved from 11% fitness in 1922 to 94% in 2025, while constitutional formalism collapsed from 89% to 3%.

The 2024–2025 period provided additional empirical evidence consistent with the theoretical framework: a Congress that couldn't reject 3 DNUs in 17 years suddenly rejected 6 in 6 months and overrode presidential vetoes with supermajorities not seen in decades, but only when these measures threatened state spending. This selective activation, with 0.91 correlation to fiscal impact, proves the system hasn't failed but evolved to protect the parasite rather than the host.

The computational tools developed and released as open-source—JurisRank, RootFinder, and Legal-Memespace—transform legal analysis from interpretation to measurement. They reveal the precise coordinates of institutional failure ([0.31, 0.89, 0.45, 0.67] at Peralta), the genealogical paths of infection (89% tracing to Peralta), and the competitive dynamics ensuring continued degradation.

For Argentina, the implications are clear but sobering. The system faces structural constraints that make reform through existing mechanisms highly improbable. Every attempt at constitutional restoration strengthens the emergency doctrine by providing new justifications for resistance. The Monte Carlo simulations suggest 82% probability of complete formalist extinction by 2030, with only 3% probability of even partial recovery.

Yet this darkness illuminates paths forward for other systems. By quantifying the infection process, it becomes possible to design resistant institutions. By mapping the parasite's genome, immunity can be engineered. By measuring degradation in real-time, intervention before metastasis becomes irreversible.

The extended phenotype of law [23] is no longer a metaphor but a measured reality. The parasite has been quantified, its genome sequenced, and its trajectory mapped. The question isn't whether Argentina can be cured; the mathematics say it cannot without revolutionary intervention that destroys both host and parasite. The question is whether other legal systems can learn from Argentina's measured transformation.

The Argentine legal system has become its own extended phenotype, building the structures that ensure its continued degradation. The parasite hasn't just won; it has rewritten the rules of the game to make its victory permanent. The available evidence suggests that constitutional mechanisms now systematically constrain constitutional governance.

The measurement is complete. The trajectory is clear. The metamorphosis isn't just continuing; it has achieved its final form: a self-reinforcing system that uses constitutional mechanisms to prevent constitutional governance.

Ethical Statement

This study does not contain any studies with human or animal subjects performed by any of the authors.

Conflicts of Interest

The author declares that he has no conflicts of interest to this work.

Data Availability Statement

The data and code that support the findings of this study are openly available in the GitHub repository "peralta-metamorphosis" at https://github.com/adrianlerer/peralta-metamorphosis.

Author Contribution Statement

Ignacio Adrián Lerer: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

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Appendices

Complete technical appendices available at: https://github.com/adrianlerer/peralta-metamorphosis

Appendix A: Algorithm Specifications (Summary) JurisRank adapts PageRank with temporal constraints: $F(i) = (1 - d)/N + d \times \Sigma(j \in C(i))[F(j) \times W(j, i) \times H(j) \times T(j, i)]$ RootFinder implements ABAN tracing genealogical paths with 76% average fidelity. Legal-Memespace maps doctrines in 4D space using PCA. Full specifications at the repository.

Appendix B: Dataset Description (Summary) 72 Argentine Supreme Court cases (1989-2025), inter-coder reliability: Cohen's Kappa = 0.83. Complete dataset at the repository.

Appendix C: Statistical Validation (Summary) Robustness confirmed: base model (τ = -0.89***), K-fold accuracy: 73.2%. Full validation at the repository.

Appendix D: Replication Instructions (Summary) Install: pip install -r requirements.txt. Run: python analysis/reproduce_paper.py. Complete tutorial at the repository.

Appendix E: 2024–2025 Congressional Dataset (Summary) Analysis of 28 DNUs: 72.7% rejection for spending-related vs 5.9% others. Correlation r = 0.91***. Full data at the repository.